ExoTransit-Net: The Critical Role of Preprocessing in Exoplanet Transit Detection via 1D-CNN

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Abstract

Automated exoplanet detection from stellar light curves is a fundamental challenge in modern astronomy, complicated by severe class imbalance and stellar noise. We present ExoTransit-Net, a 1D convolutional neural network pipeline for detecting planetary transits in Kepler space telescope light curves. Using the Kepler labelled time series dataset (5,087 stars, 37 confirmed planet hosts), we systematically evaluate the impact of preprocessing on detection performance. Our key finding is that Savitzky-Golay detrending reduces false alarms by 94% (143 to 8) and improves F1 score by 186% (0.05 to 0.143) compared to augmentation alone, demonstrating that preprocessing quality dominates model performance in rare astronomical event detection.

1 Introduction

Exoplanet detection via the transit method relies on identifying periodic dips in stellar brightness caused by planets passing in front of their host stars. The Kepler space telescope observed over 150,000 stars for four years, producing light curves with approximately 70,000 flux measurements per star. Manual inspection of this volume of data is infeasible, motivating automated machine learning approaches.

The primary challenge is severe class imbalance — in our dataset, only 37 of 5,087 stars (0.7%) host confirmed exoplanets. Additionally, stellar variability creates slow brightness oscillations that can be confused with transit signals, leading to false positives. Standard accuracy metrics are misleading in this regime; a model predicting "no planet" for every star achieves 99.3% accuracy while finding zero planets.

In this work we present ExoTransit-Net, systematically evaluating four preprocessing strategies on a 1D-CNN architecture. Our key finding is that Savitzky-Golay detrending — which removes slow stellar trends while preserving short-duration transit signals — produces the best balance of precision and recall.

2 Related Work

Shallue & Vanderburg (2018) demonstrated that deep neural networks can identify exoplanet candidates from Kepler light curves, achieving 98.8% accuracy using a CNN on phase-folded light curves. However, their approach requires prior knowledge of orbital period for phase folding. Our approach operates directly on raw light curves without requiring period estimation, making it applicable to initial candidate screening.

3 Dataset

We use the Kepler labelled time series dataset containing 5,087 stars with 3,197 flux measurements each, split into 5,087 training and 570 test stars. Labels indicate confirmed exoplanet host (37 stars, label 2) or non-host (5,050 stars, label 1), converted to binary 0/1.

The severe class imbalance (1:136 ratio) makes standard training ineffective. We address this through class weighting and data augmentation — generating 11 versions of each planet star (noise addition, time shifting, amplitude scaling), expanding planet examples from 37 to 407 and reducing the ratio to approximately 1:10.

4 Methodology

Our 1D-CNN architecture consists of three convolutional blocks, each containing a Conv1D layer, MaxPooling1D, BatchNormalization and Dropout, followed by a Dense classifier with sigmoid output. The model has 491,073 parameters.

Table 1: Model architecture

Layer	Output Shape	Parameters
Conv1D (32, k=11)	(3187, 32)	384
MaxPool + BN	(796, 32)	128
Conv1D (64, k=11)	(786, 64)	22,592
MaxPool + BN	(196, 64)	256
Conv1D (128, k=11)	(186, 128)	90,240
MaxPool + BN	(46, 128)	512
Dense (64) + Dropout	(64,)	376,896
Dense (1, sigmoid)	(1,)	65

We evaluate four preprocessing configurations:

1. Standardization only (baseline)
2. Standardization + class weighting (15×) + augmentation
3. Uniform filter detrending (window=200) + augmentation
4. Savitzky-Golay detrending (window=201, order=2) + augmentation

5 Results

Table 2: Model performance across preprocessing strategies

Configuration	Precision	Recall	F1	False Alarms
Baseline (136× weight)	0.01	1.00	0.02	565
Augmentation (15× weight)	0.03	0.80	0.05	143
Uniform filter detrending	0.00	0.00	0.00	0
SavGol detrending	0.11	0.20	0.14	8

The baseline model with aggressive class weighting (136×) achieved perfect recall but 565 false alarms — predicting "planet" for every star. Reducing class weight to 15× with augmentation improved precision but still produced 143 false alarms.

Naive uniform filter detrending removed transit signals along with stellar trends, reducing all metrics to zero. Savitzky-Golay detrending, which fits local polynomials rather than computing simple averages, preserved transit-scale signals while removing slow stellar variability, achieving F1=0.143 with only 8 false alarms.

6 Discussion

Our results demonstrate that preprocessing quality is the dominant factor in exoplanet transit detection performance, more so than model architecture or training strategy. The 94% reduction in false alarms from augmentation-only to SavGol detrending highlights the importance of domain-specific signal processing.

The primary limitation is the extremely small positive class — 37 planet stars is insufficient for robust generalization even with augmentation. Future work should explore phase folding to amplify the transit signal, LSTM or attention-based architectures better suited to periodic patterns, and larger datasets from TESS mission data.

7 Conclusion

We presented ExoTransit-Net, demonstrating that Savitzky-Golay detrending reduces false alarms by 94% and improves F1 by 186% over augmentation alone. Our systematic evaluation across four preprocessing strategies establishes that signal processing quality dominates model performance in rare astronomical event detection, with implications for other imbalanced time-series classification problems in astrophysics.

References

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